



sAlfer Lab

Joint lab on Safety and Security of AI

Computer Vision Technologies and Biometrics

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**M.Sc. Degree in Computer Engineering, CyberSecurity and
Artificial Intelligence**



University
of Cagliari, Italy

Department of Electrical
and Electronic
Engineering





<https://www.saiferlab.ai>

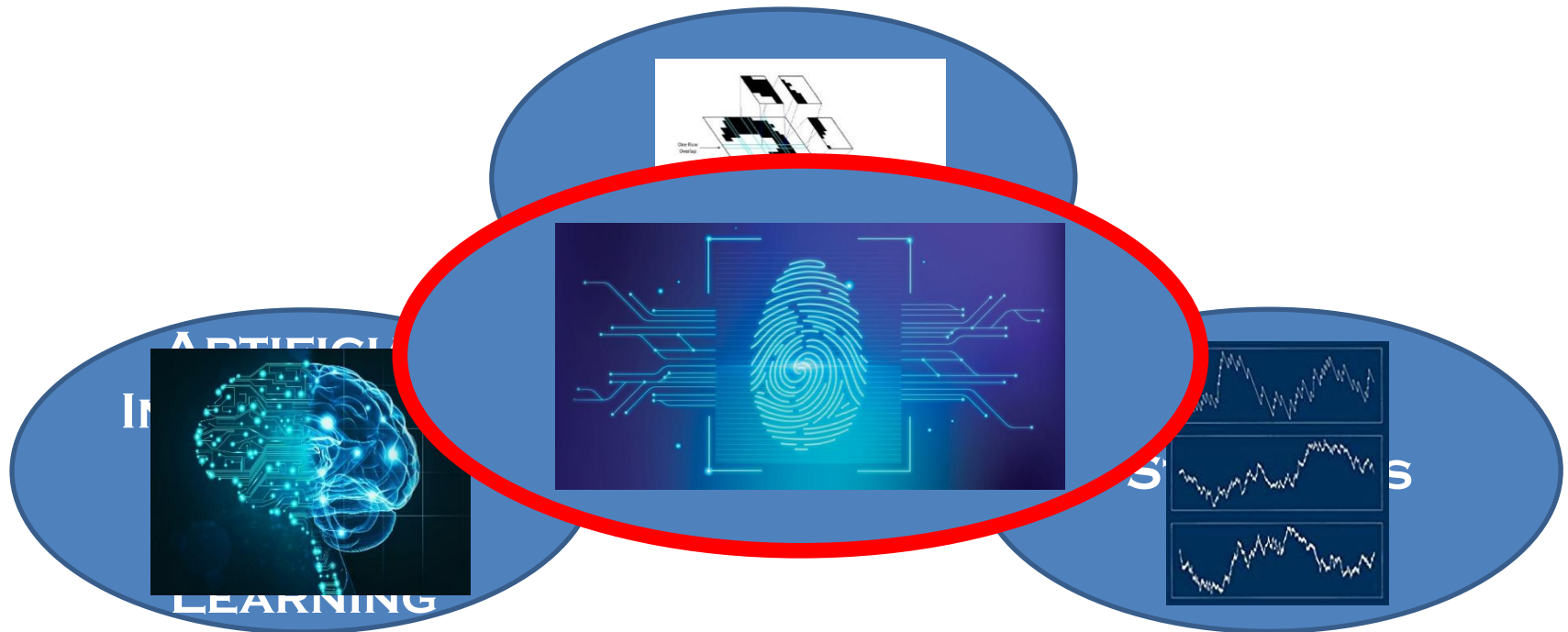
Overview of the course

- **Computer vision fundamentals**
 - Definition
 - Image formation and processing
 - Image alignment and stitching
 - Depth and motion estimation
- **Biometric technologies**
 - Definition
 - Fingerprints
 - Faces
 - Multiple biometrics
 - Other biometrics
- **Security and biometrics**
 - Vulnerabilities of biometric systems
 - Presentation Attacks Detection
 - Design of countermeasures against spoofing attacks
- **Behavioural security**
 - Crowd analysis
 - Anomalous behaviour
 - Human-in-the-loop

«Fusion» of different disciplines



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What will you learn during lectures?



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- Basics of computer vision systems design and evaluation
- Main modules of a biometric system
- How pattern recognition and machine learning concepts are applied to computer vision and biometric recognition
- Design of presentation attacks detectors as a basic protection «infrastructures» for biometric systems
- How people behavior can be «detected» for mass-oriented video-surveillance systems

What will you learn to do during the laboratory exercises?



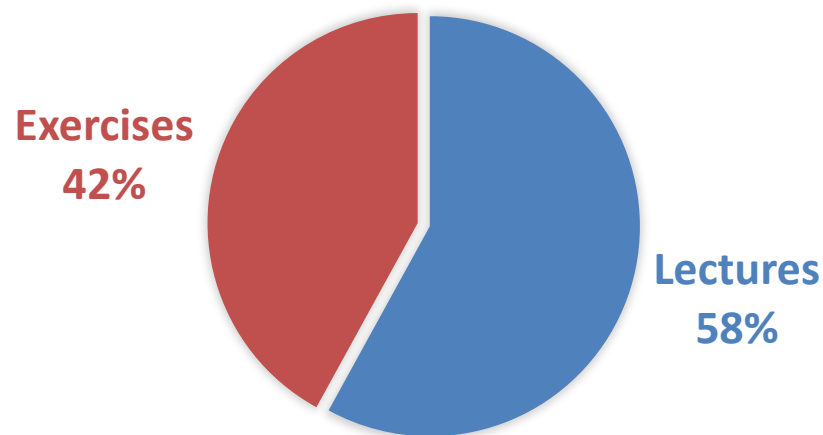
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- **To design and test computer vision and biometric recognition systems**
- **Using advanced programming languages and tools for computer vision and biometric applications**
 - Matlab/Python/C
 - Image processing
 - Pattern recognition
 - Deep learning
- **Exploring pros and cons of video-based analysis**

Course organization



- **Lectures and laboratory exercises**
 - Wednesday, 11.00am, I_IB room (3 hours)
 - Thursday, 9.00am, I_IB room (3 hours)
- **Balance between lectures and exercises**



How to gain the exam score?



First step: project assignment

- Assignment is done one week before the official exam date
- Same size/difficulty of the ones given during the laboratory exercises
- This must be carried out alone (no groups)
- Each assignment must be completed by one week

Second step: oral verification on the whole course program

- The day of the official exam date
- 30 minutes of interview with me and my co-workers
- Detailed description of your project by slides presentation

Cross your fingers and good luck! 😊

Some M.Sc. thesis topics



- Latent fingerprints and verification abilities
- 3D Face reconstruction by laser scanner and images
- Defenses against presentation attacks
- Generation of synthetic crowds: simulation and analysis
- Embedding audio-messages in a targeted user's voice
- Object detection and tracking under various scenarios

How to study



- **Follow the course**
- **Do questions**
 - During lecture/lab exercises
 - During the break
 - By e-mail
- **Download the course slides**
- **Integrate your notes and book information (no wikipedia)**
- **Do (many) programming exercises at home**

MS Teams

- <https://teams.microsoft.com/1/team/19%3Ad8e28d32541c4014b14d247934b24d7b%40thread.tacv2/conversations?groupId=a36ace60-f9e5-49d0-869e-f767e0c1462e&tenantId=6bfa74cc-fe34-4d57-97d3-97fd6e0edee1>
- Team code: 33i757z



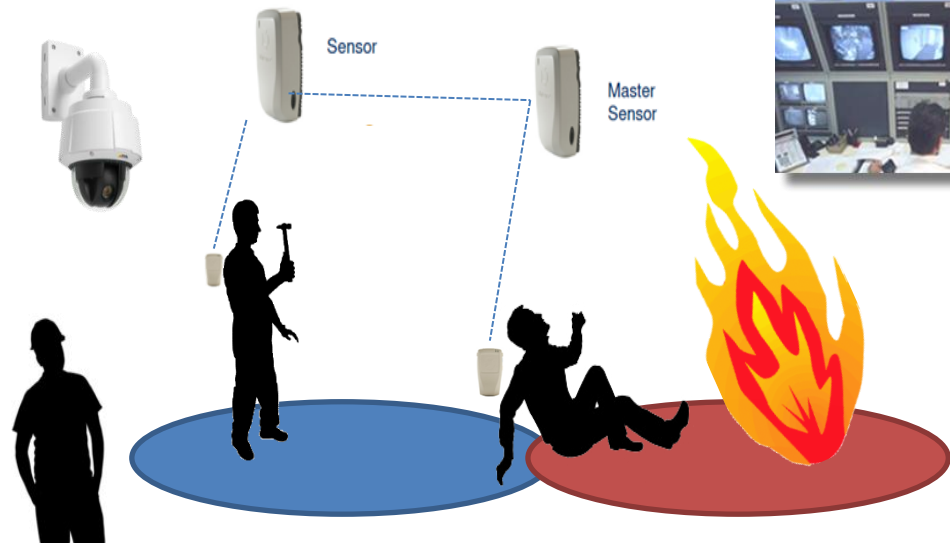
Bibliography

- R. Szeliski, **Computer Vision: Algorithms and Applications**, 2nd Edition, <https://szeliski.org/Book/>
- A. Jain et al., **Handbook of Biometrics**, Springer
- B. Bhanu and A. Kumar, **Deep learning in biometrics**, Springer
- V. Murino et al., **Group and crowd behavior for computer vision**, Academic Press
- D. Maltoni et al., **Handbook of fingerprint recognition**, Springer
- H. Liu, **Face Detection and Recognition on Mobile Devices**, Elsevier

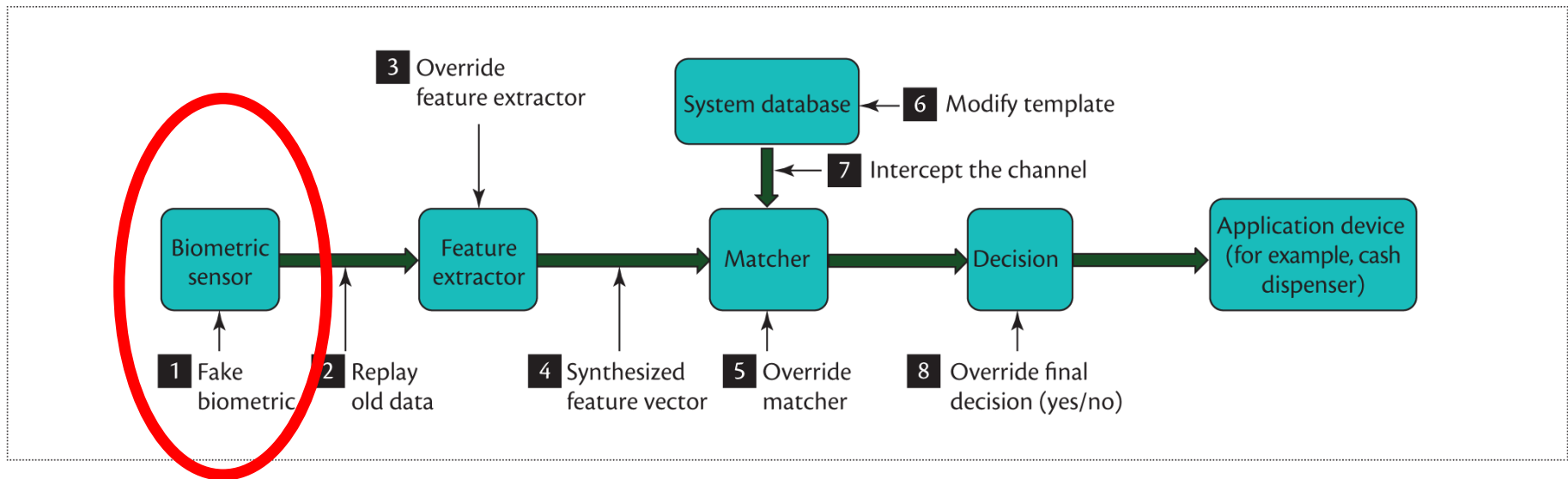
Use-inspired research



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Vulnerability points



Fingerprint anti-spoofing



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Stand-alone fingerprint
verification system
with anti-spoofing
tailored for different
capture devices



«Genuine» spoofs from our lab



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What about... deep fakes?



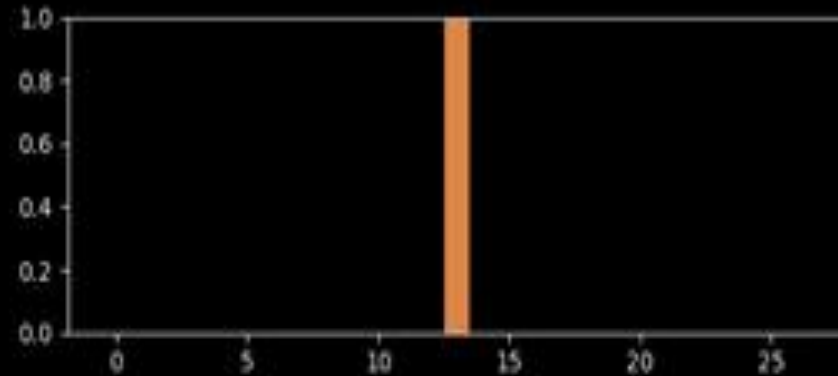
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Behavioural security: starting point



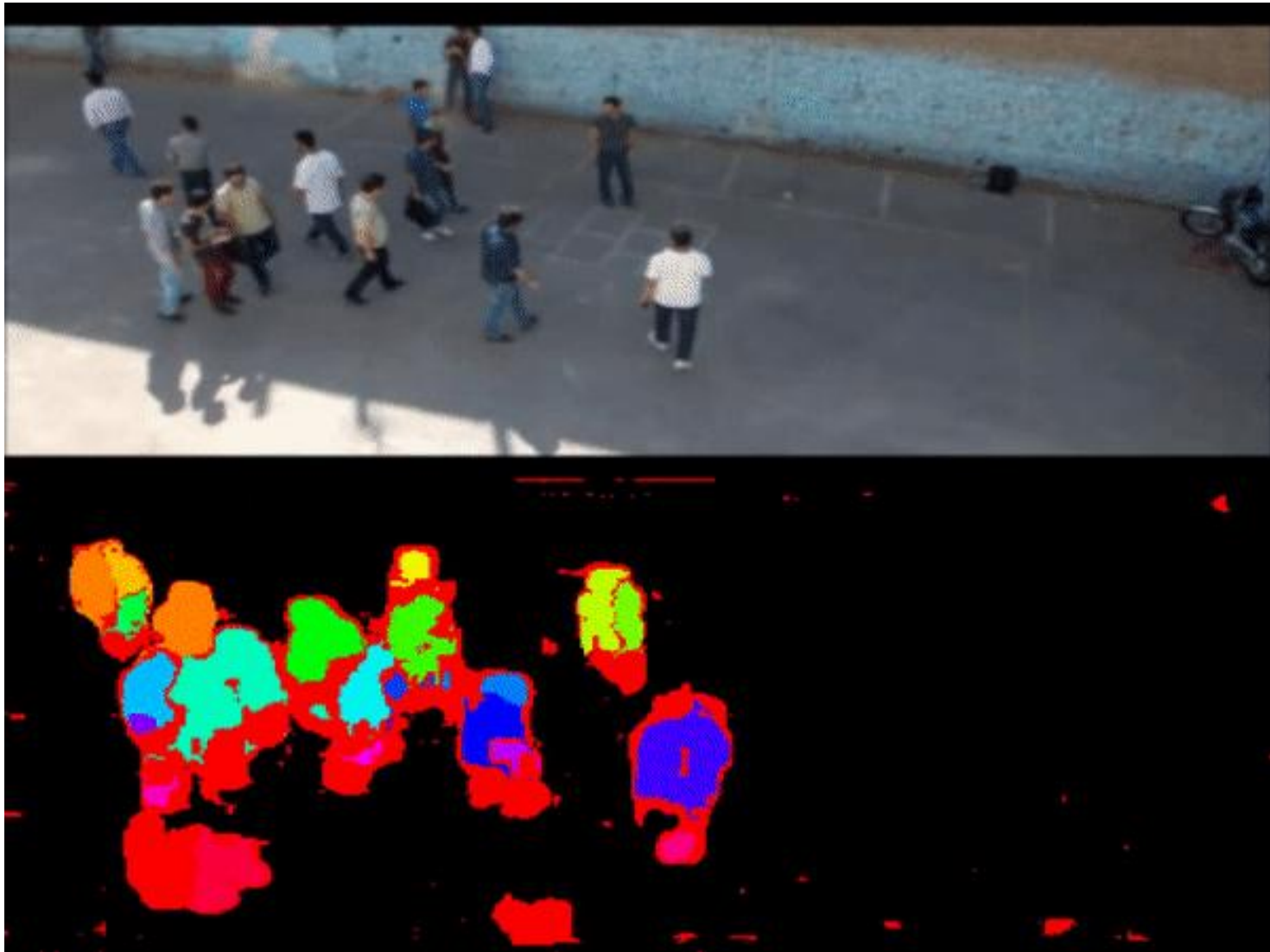
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Cluster detection



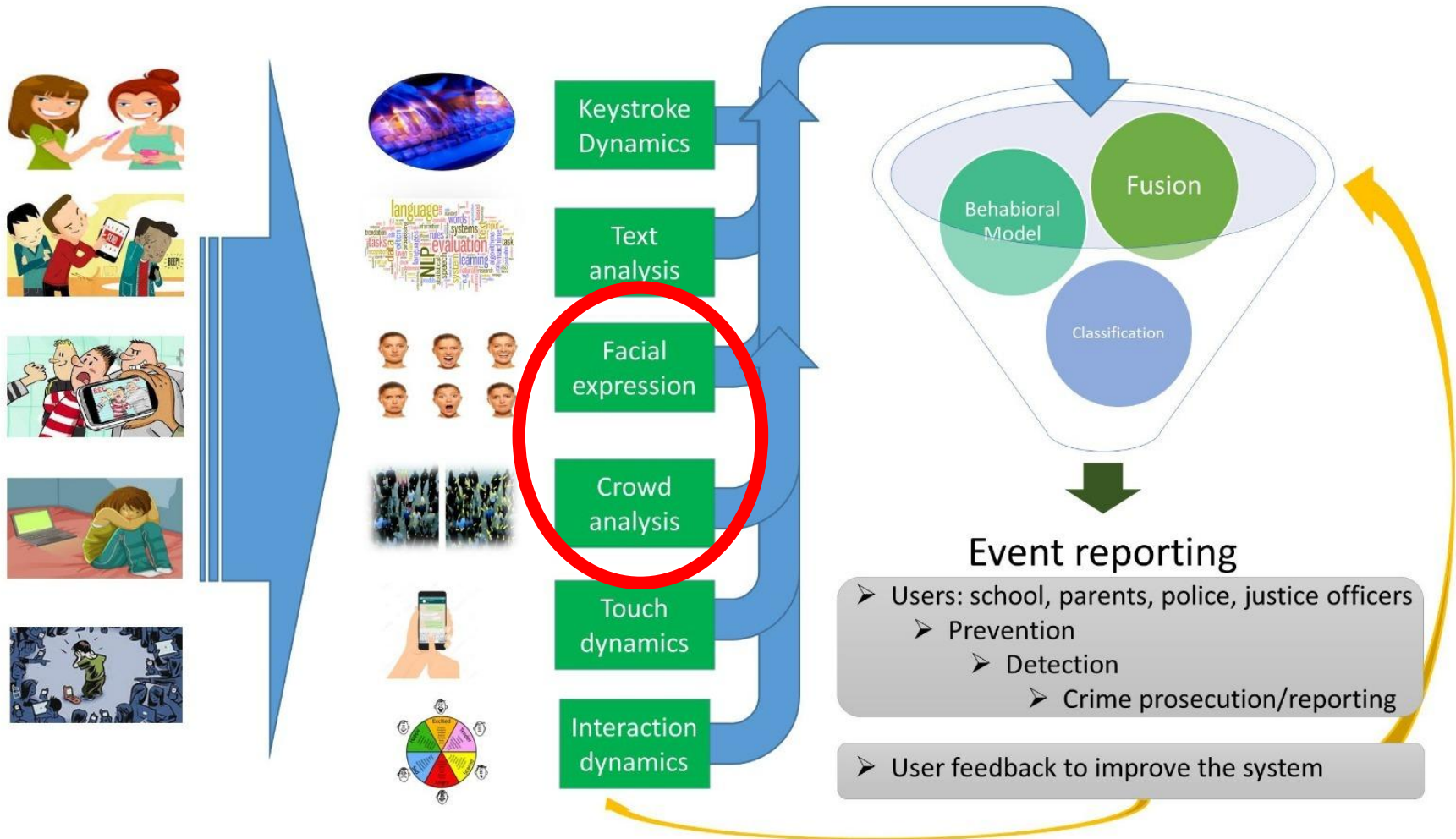
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A recent project: the BullyBuster



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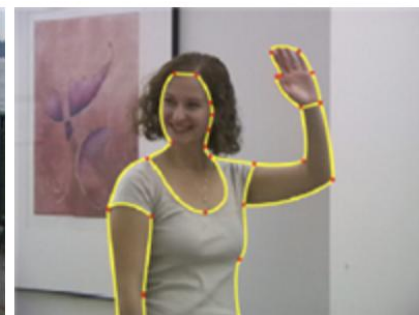
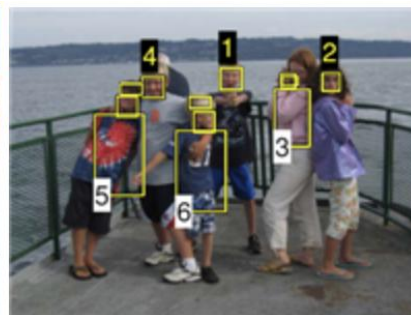
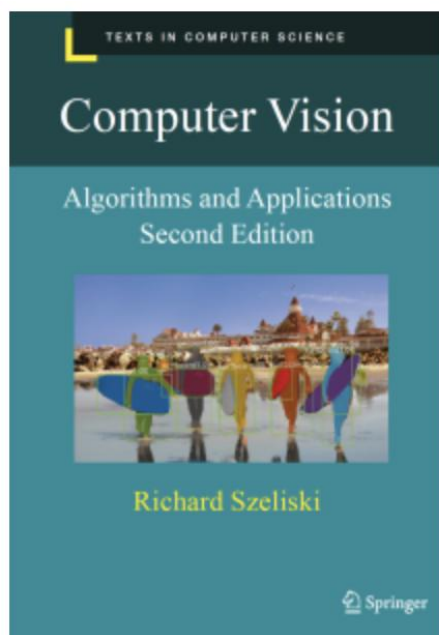


Bari – Cagliari – Foggia – Napoli «Federico II»

Let's start: some premises



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The concept of «pattern»



Many computer vision applications can be viewed as an application of pattern recognition



Pattern Recognition is the scientific discipline which studies methods and algorithms aimed to model, measure, compare and classify “patterns” automatically



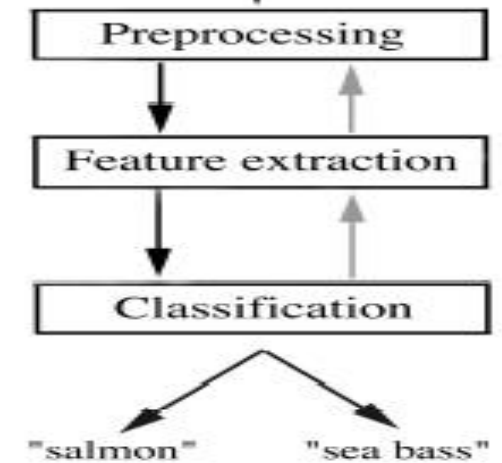
A pattern is whichever type of data (numerical, categorical,...) representing a certain physical phenomenon

Sounds, documents, persons, fingerprints, faces, ...

A conceptual example



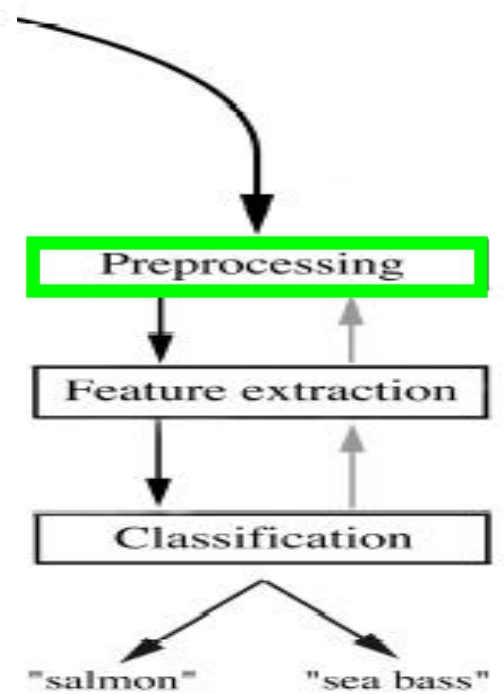
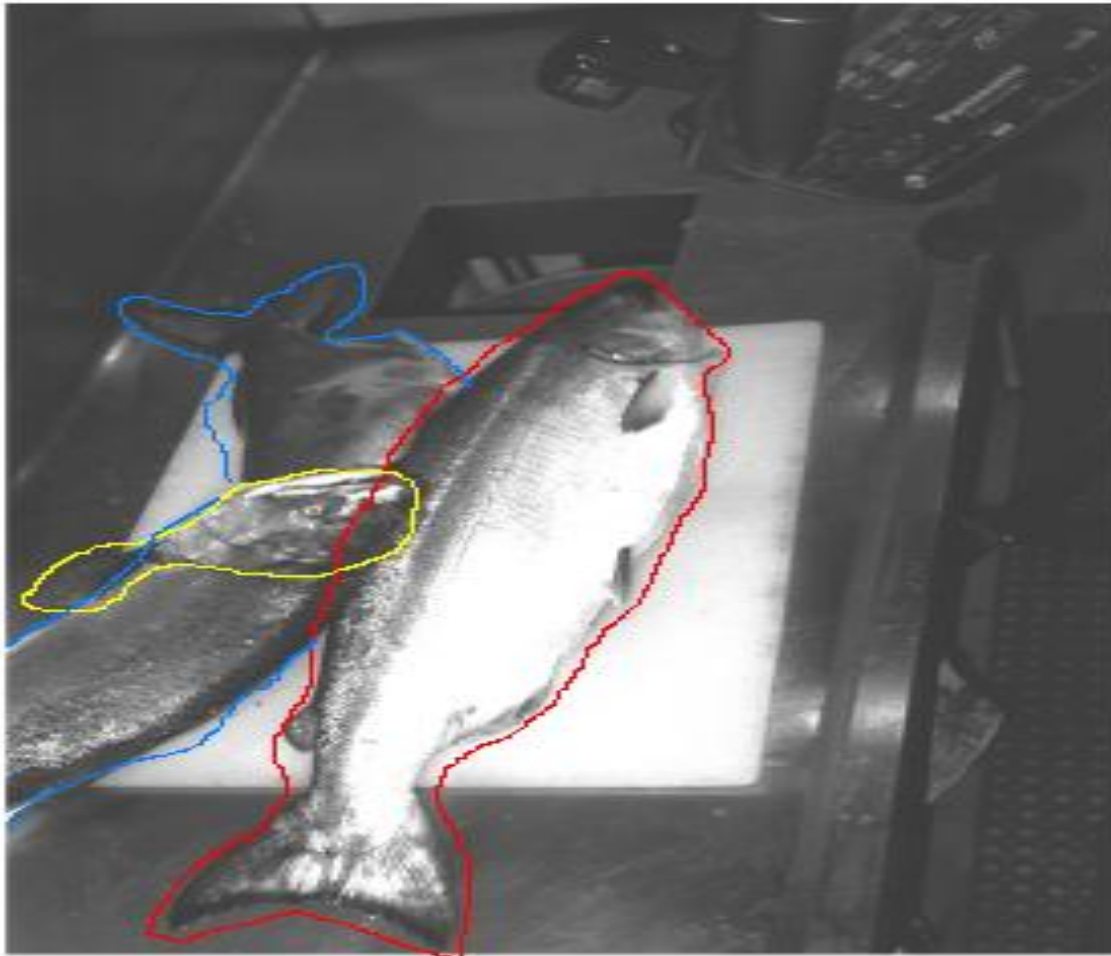
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Pre-processing



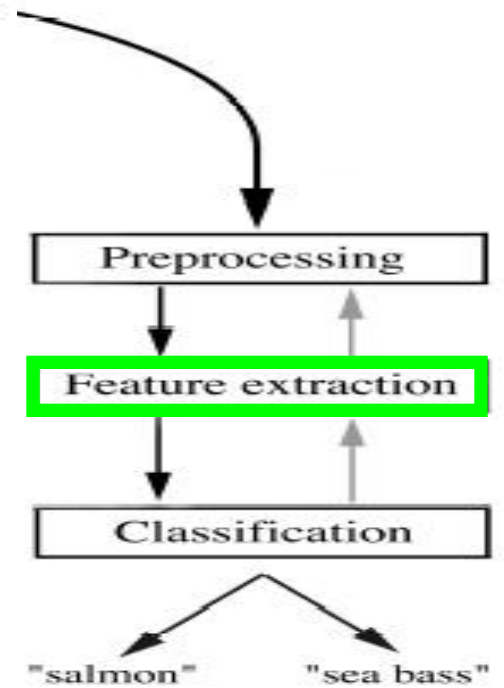
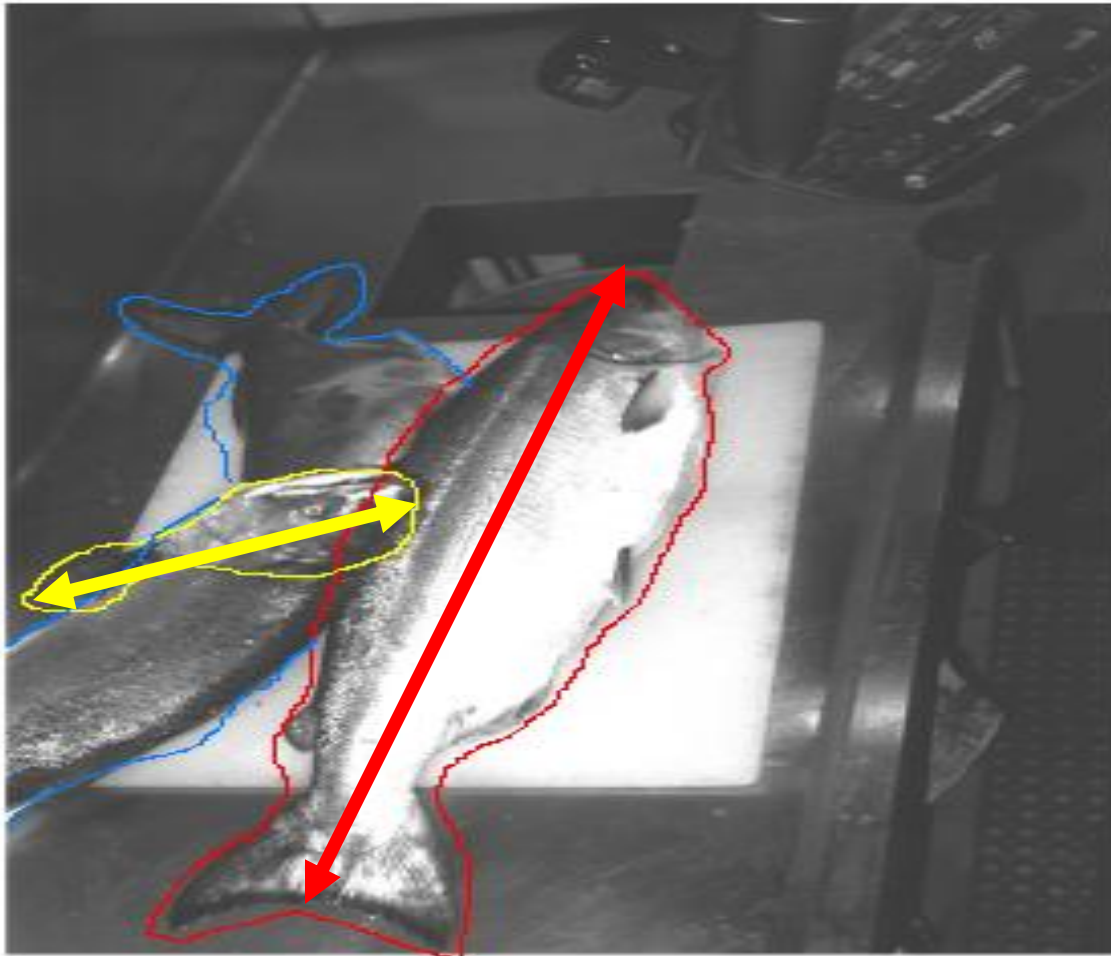
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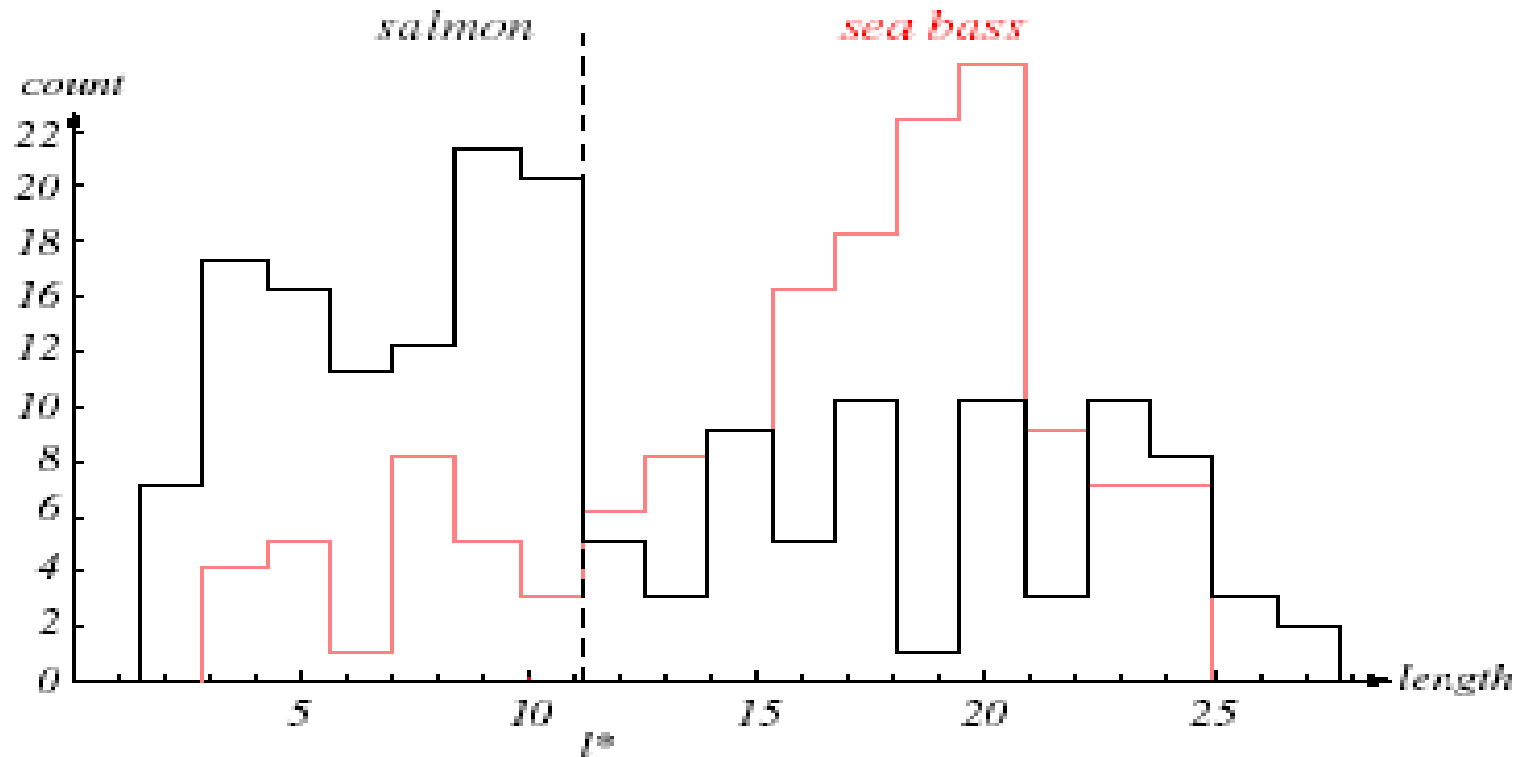
Feature extraction



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Classification

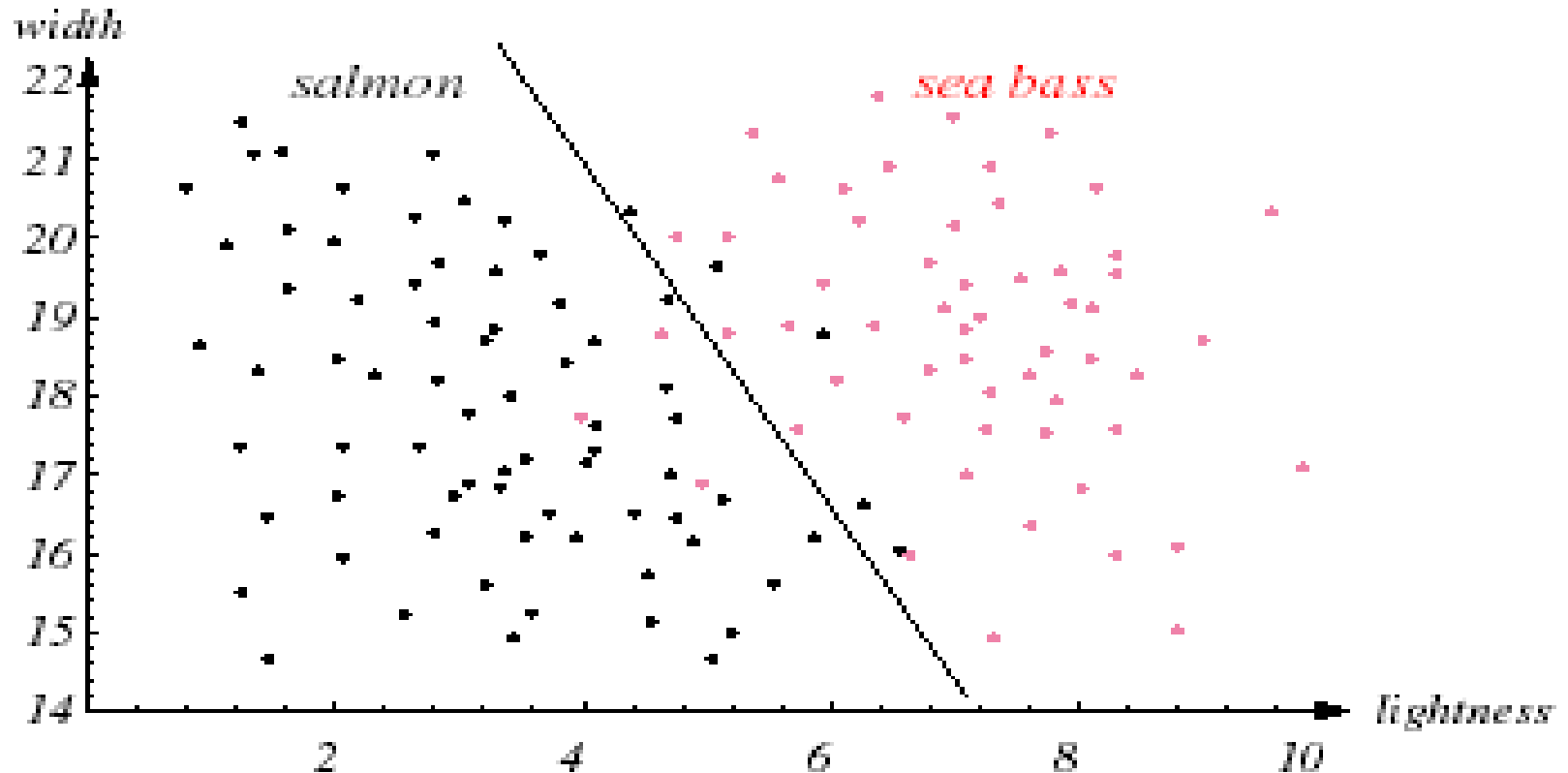


The threshold l^* is applied to “separate” salmons and sea basses

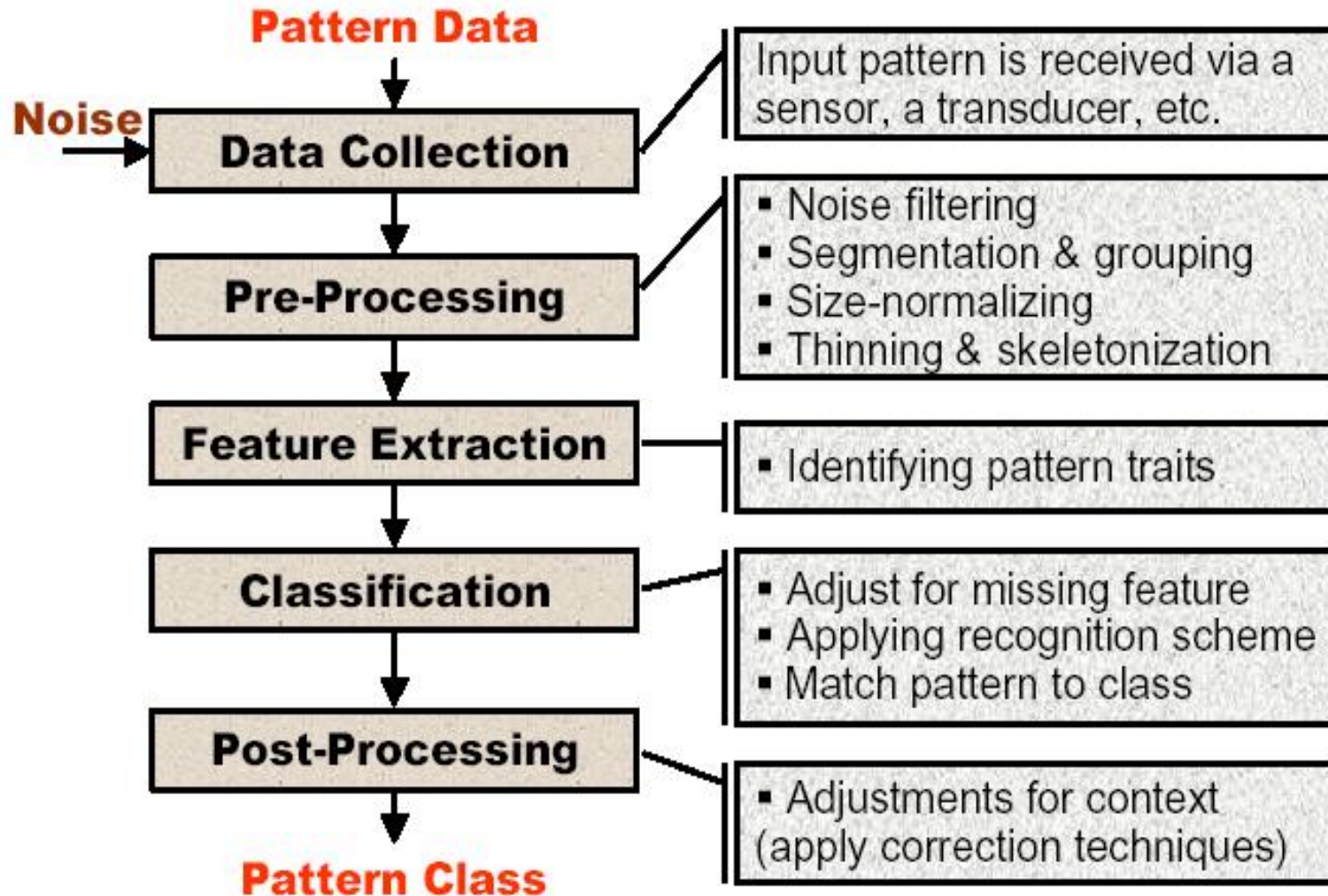
Therefore: **if** $l > l^*$ **then** fish_class $\leftarrow$ sea bass **else** fish_class $\leftarrow$ salmon

- **Pre-processing**
 - It is necessary to separate the “information” from the data
 - For example, the face image from the background in a snapshot
- **Features extraction**
 - They can be affected by noise
 - A feature can be very useful, but very difficult to measure
- **Classification**
 - Feature space projection

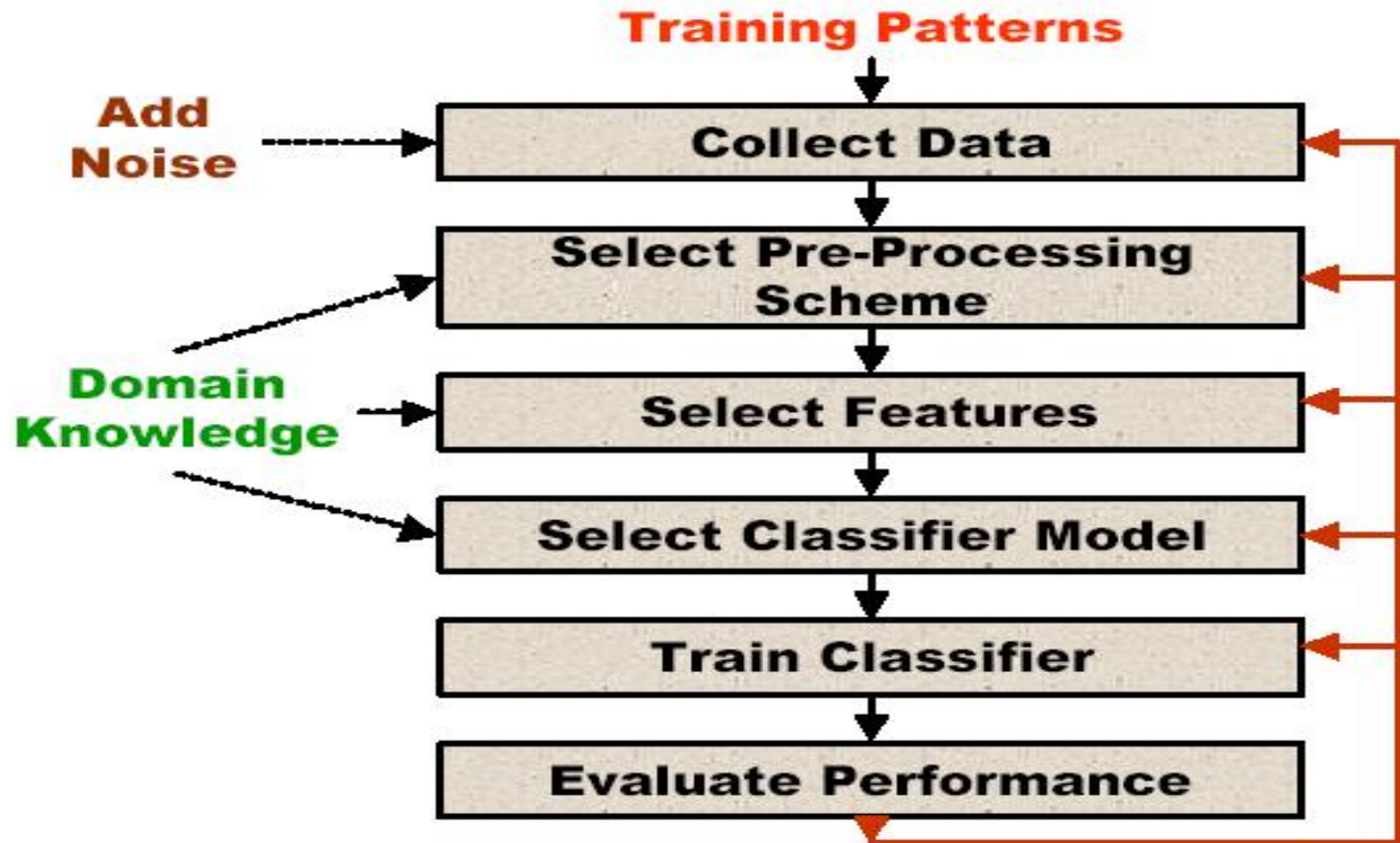
The features space



To sum up...



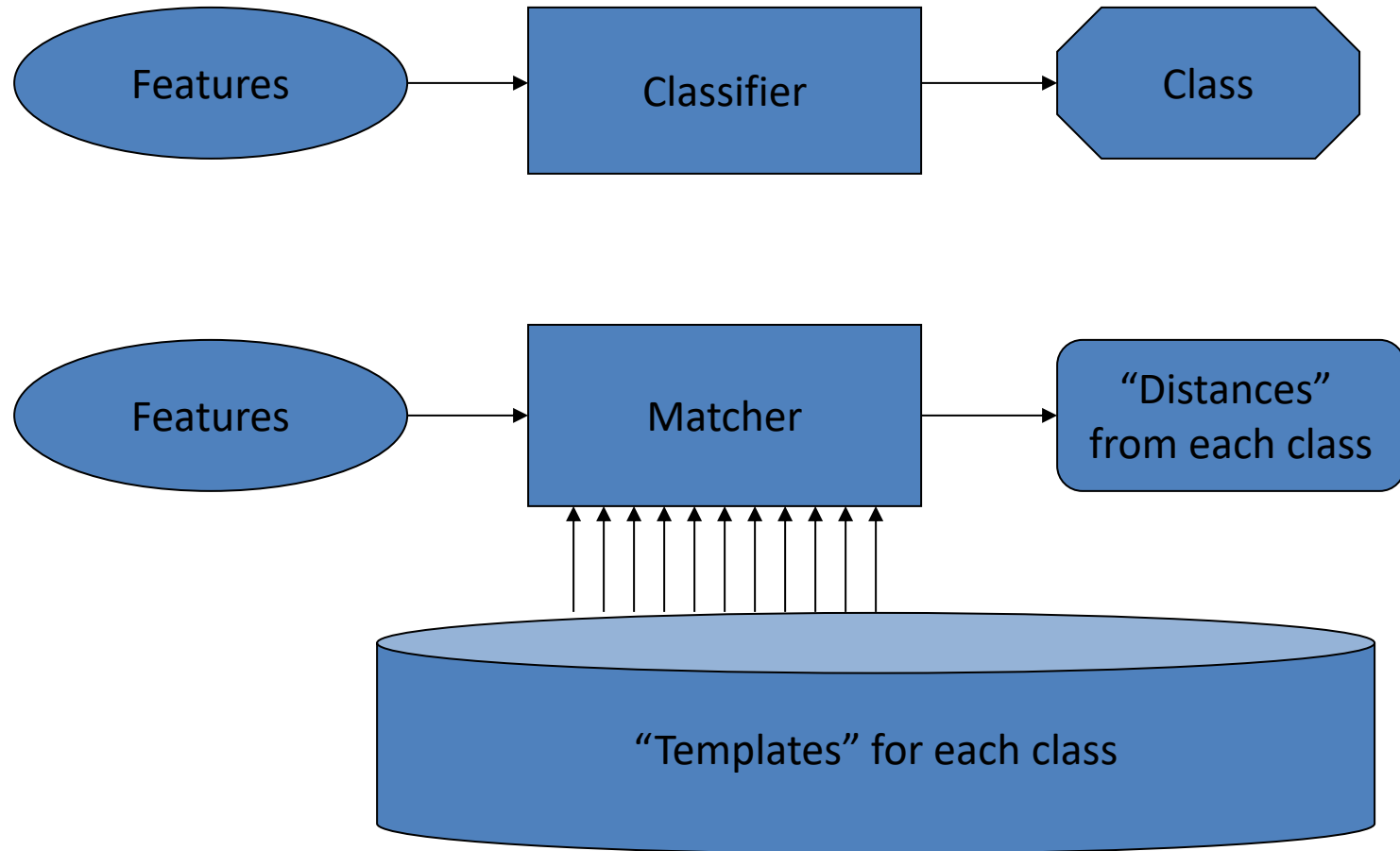
Design of a pattern recognition system



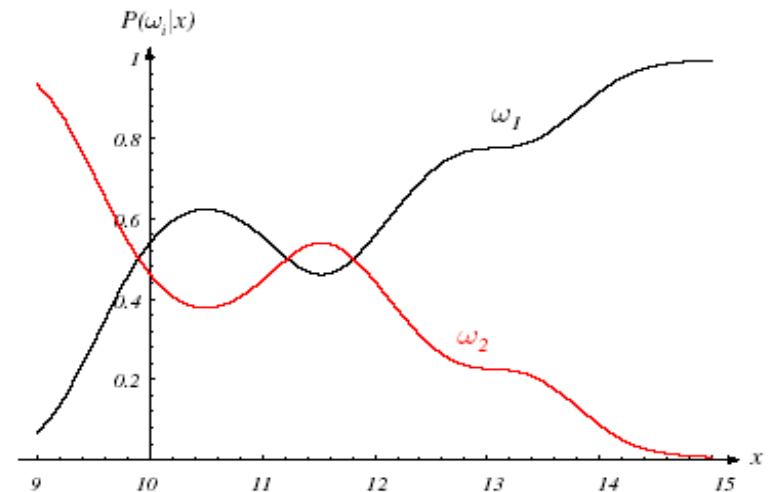
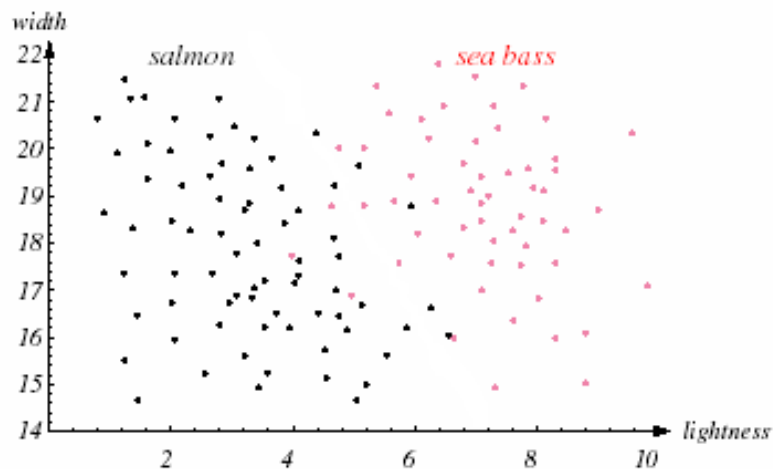


- **Vectors of**
 - Real, integers, discrete values
 - Feature space → fixed size
- **Strings**
 - Vectors of different sizes
- **Graphs**
 - The “structure” of the data is pointed out

Classification approaches



- Based on
 - Feature space geometry
 - Estimation of (a posteriori) probabilities





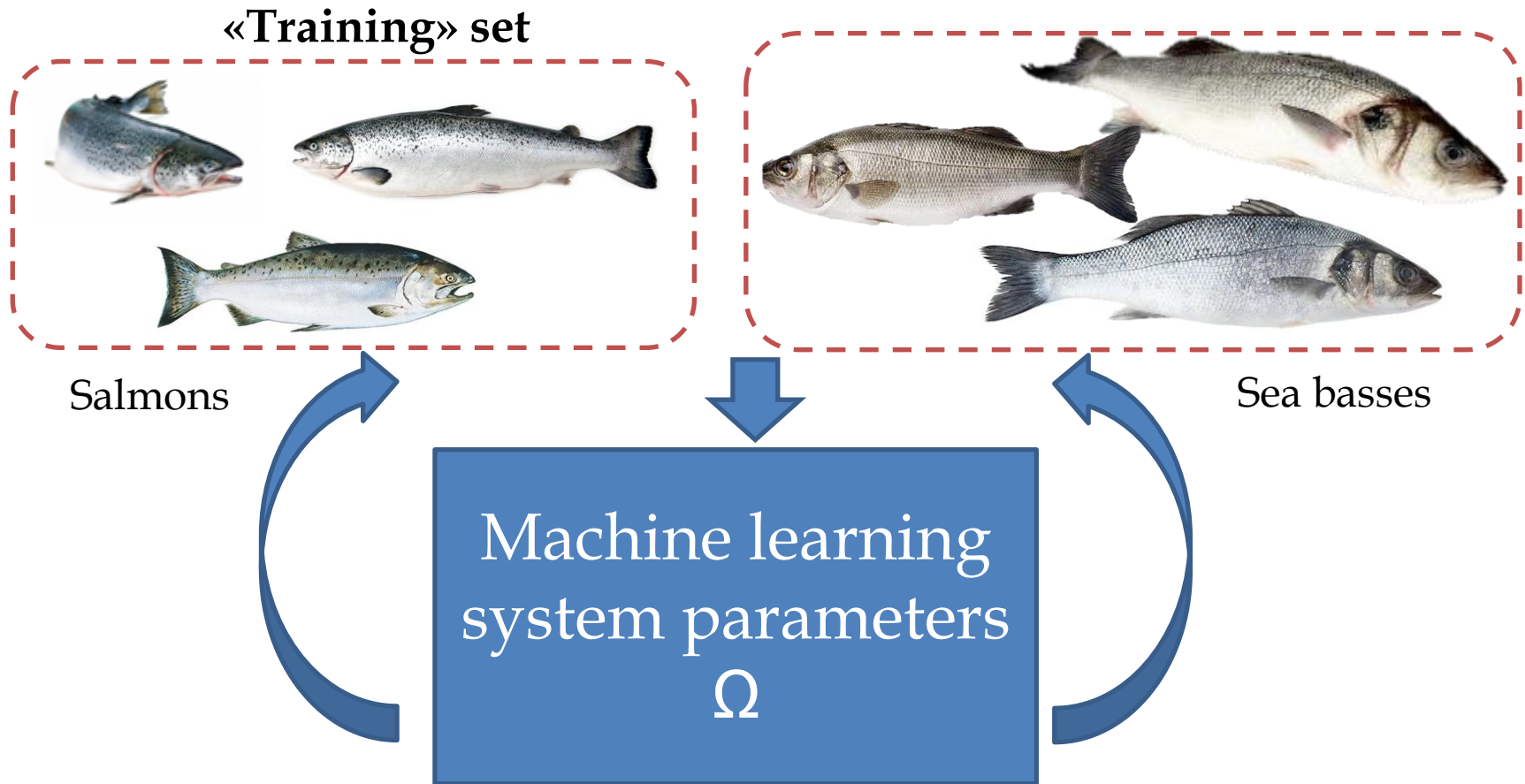
- **K-Nearest Neighbour (K-NN)**
- **Neural networks**
 - Perceptron, MultiLayer-Perceptron (MLP)
 - Recursive Neural Networks*
- **Stochastic models**
 - Hidden Markov Models (HMM)
 - Recursive Neural Networks* (RNN)
- **Support Vector Machines (SVM)**

- **Discipline of Artificial Intelligence**
 - Aimed at creating automatic machine able to simulate «human knowledge» about a certain problem
- **Key-approach: «learning» by examples**
- **Estimation of concepts as:**
 - A posteriori probability
 - «Distance»

Basics of machine learning

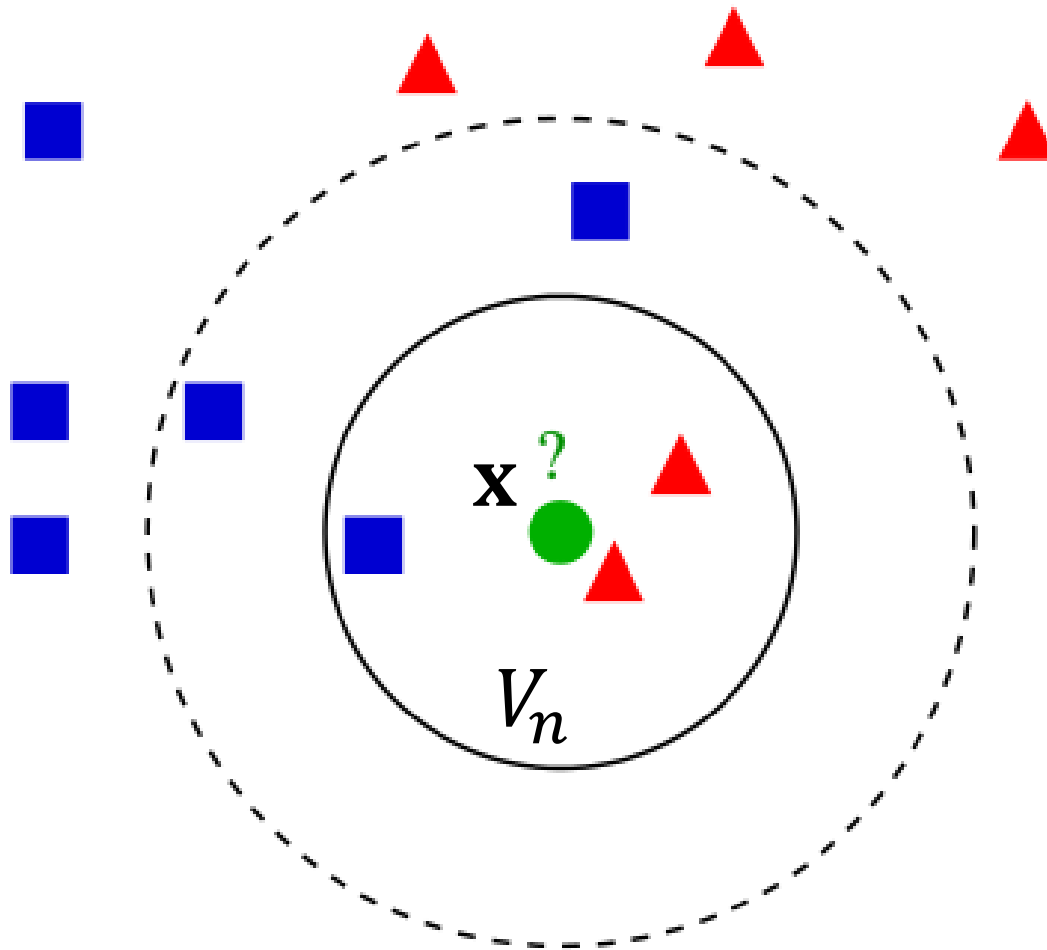


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$$P(\omega|\mathbf{x}; \Omega) = \frac{p(\mathbf{x}|\omega; \Omega)P(\omega)}{p(\mathbf{x}; \Omega)}$$

K-Nearest Neighbour



$$p_n(\mathbf{x}) = \frac{k_n/n}{V_n}$$



An SVM is a separating hyperplane in a high, non-linear feature space Z of the input.

$$f(x) = \sum_{i=1}^m \alpha_i y_i K(x_i, x)$$

$(x_i, y_i) \in X \times \{-1, 1\}, i = 1 \dots m$ is the set of i/o examples.

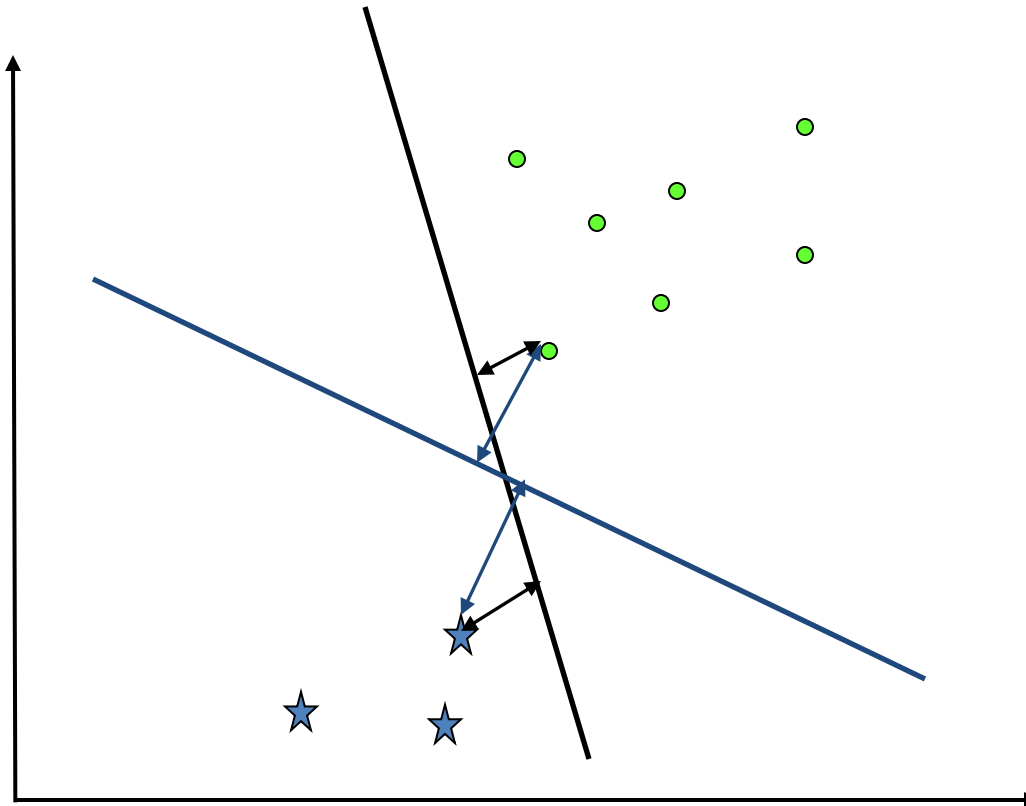
The Kernel K implicitly defines feature space Z .

The SVM is computed by maximizing the margin (or a soft margin) between the given set of examples.

A simple example



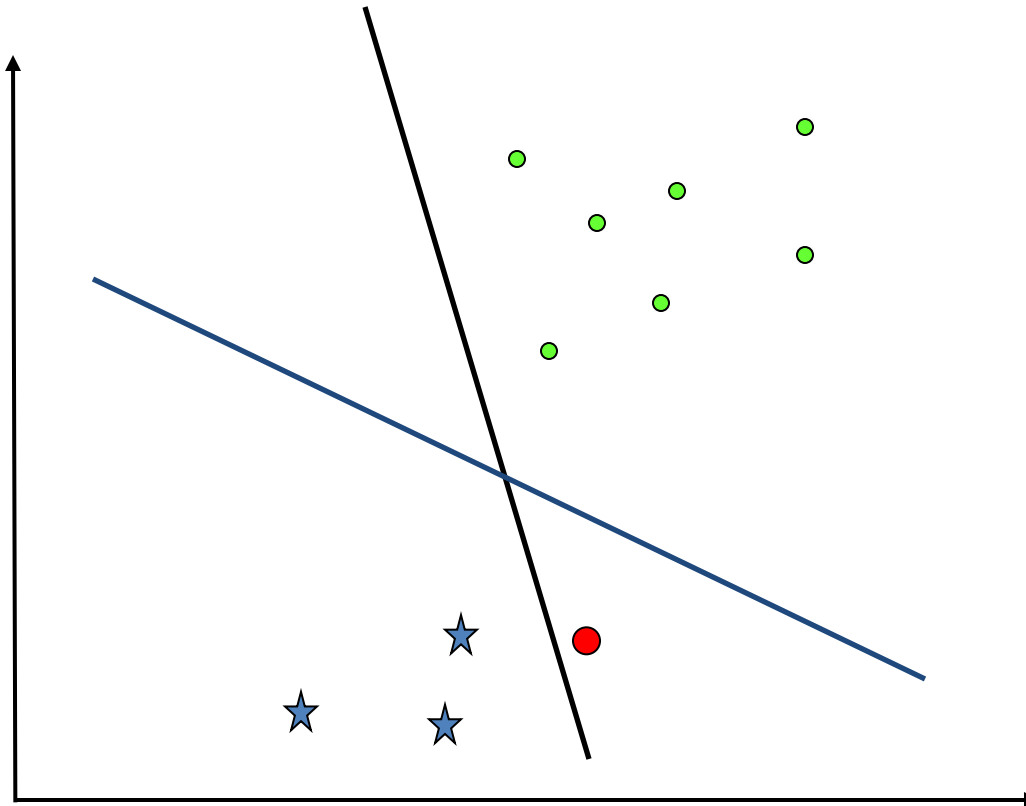
In the simplest case the SVM is a perceptron with maximum margin. This corresponds to using a **linear** kernel



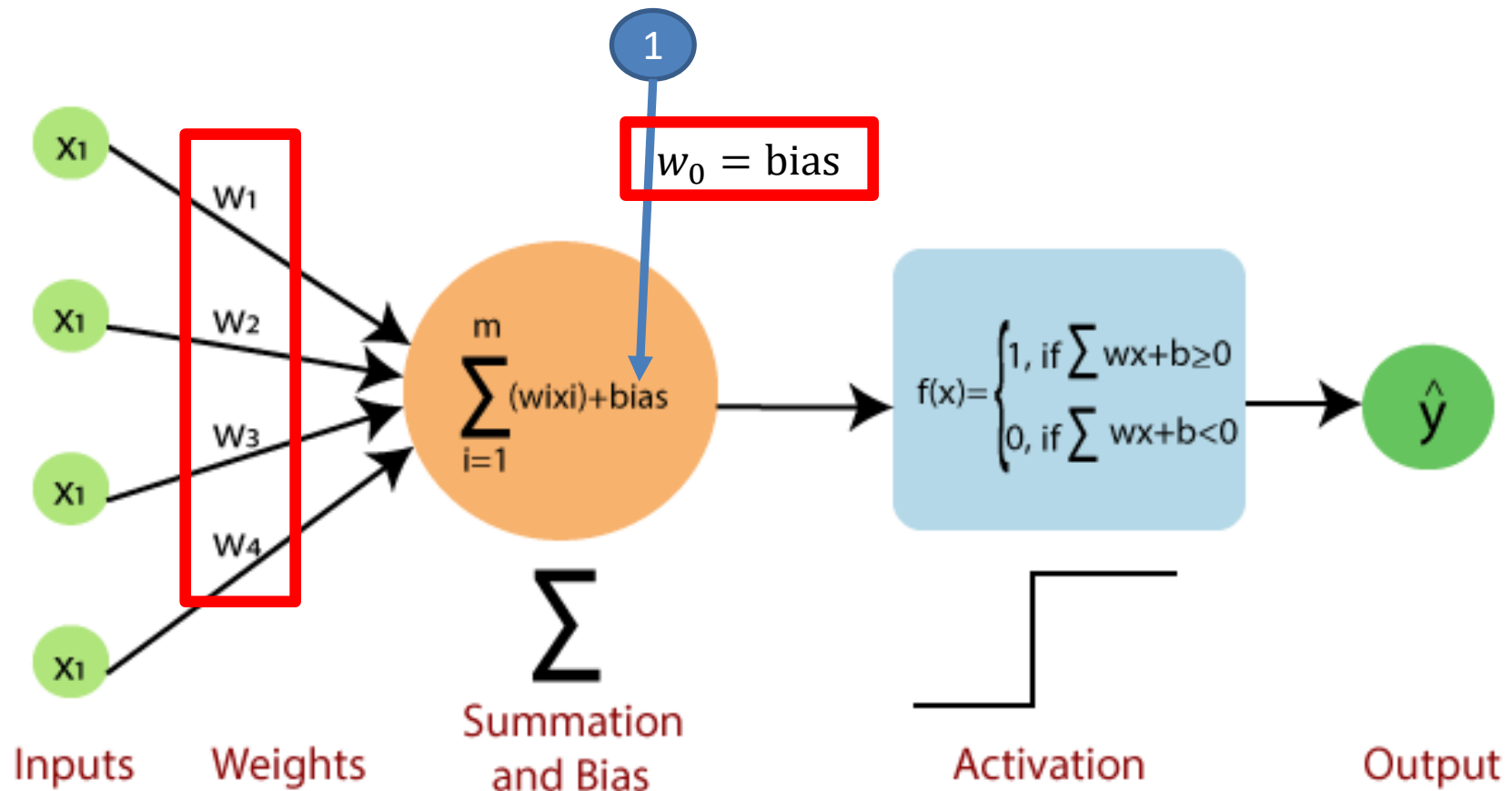
Benefits of SVMs



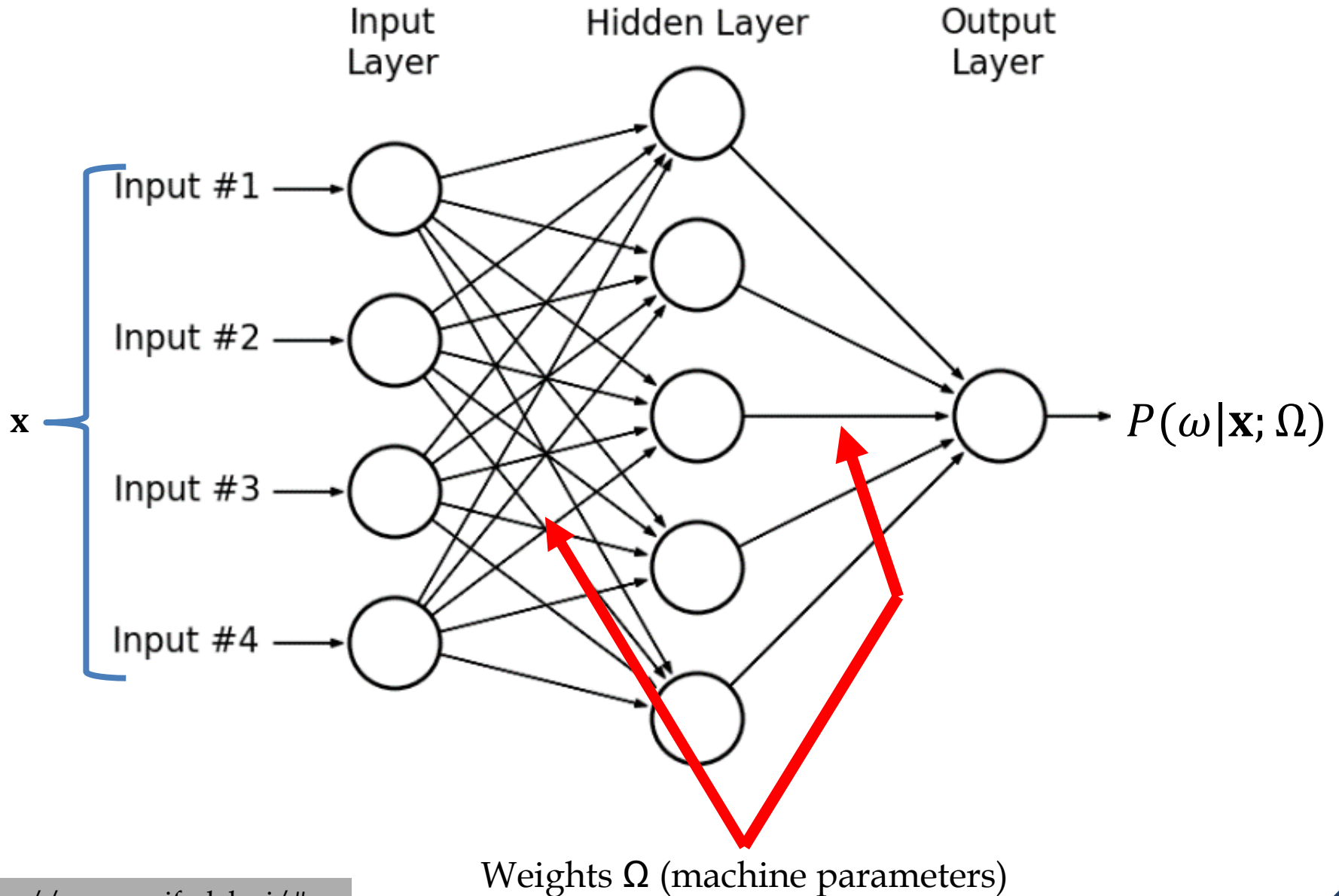
- a) Solution is **unique**
- b) Solution has **smaller** expected error



Perceptron



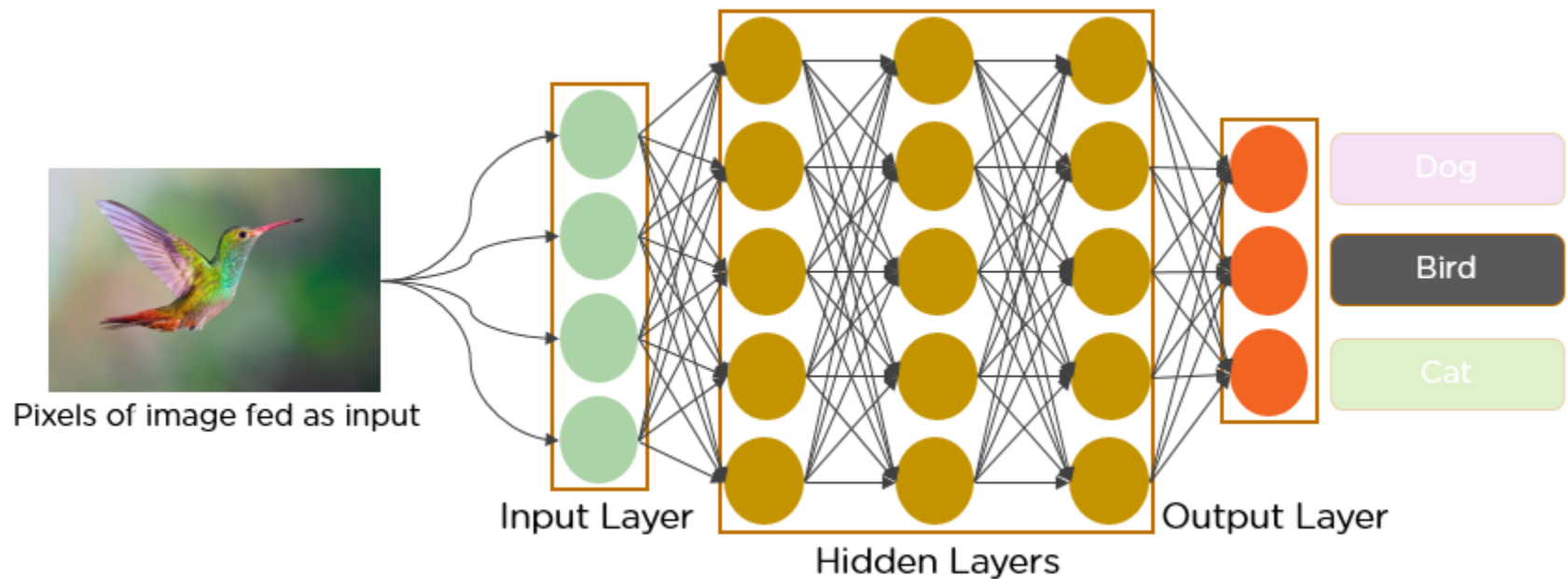
MultiLayer Perceptron (MLP)



Deep MLPs



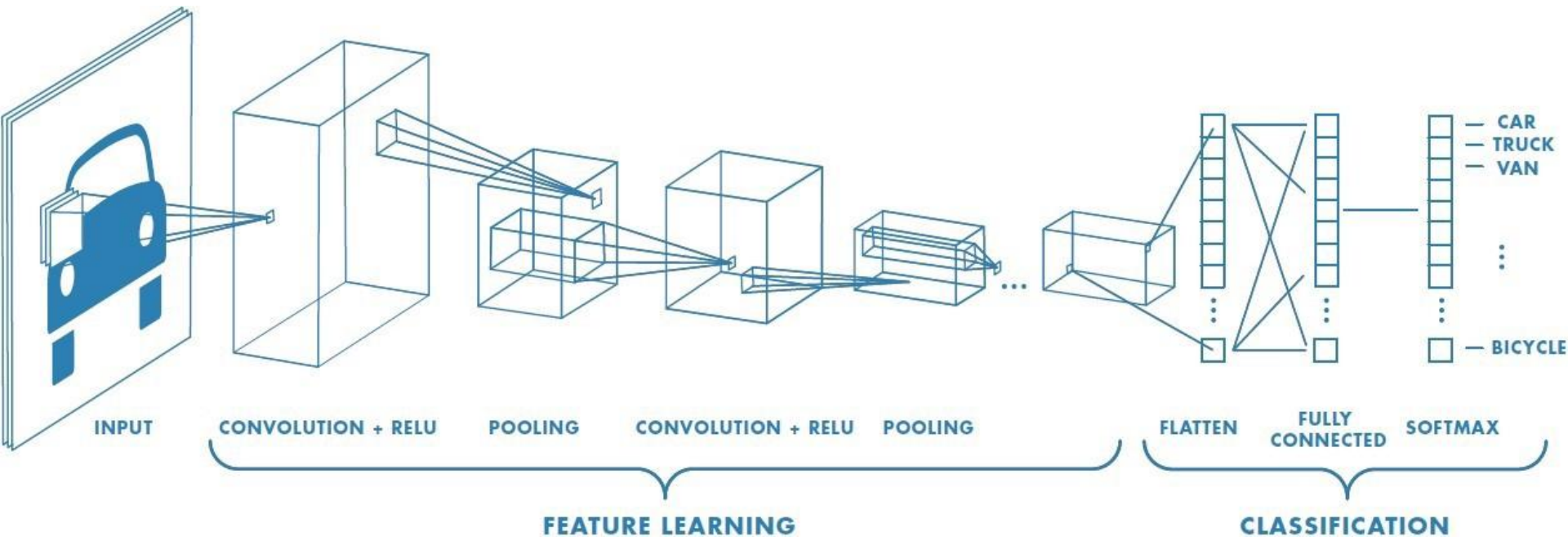
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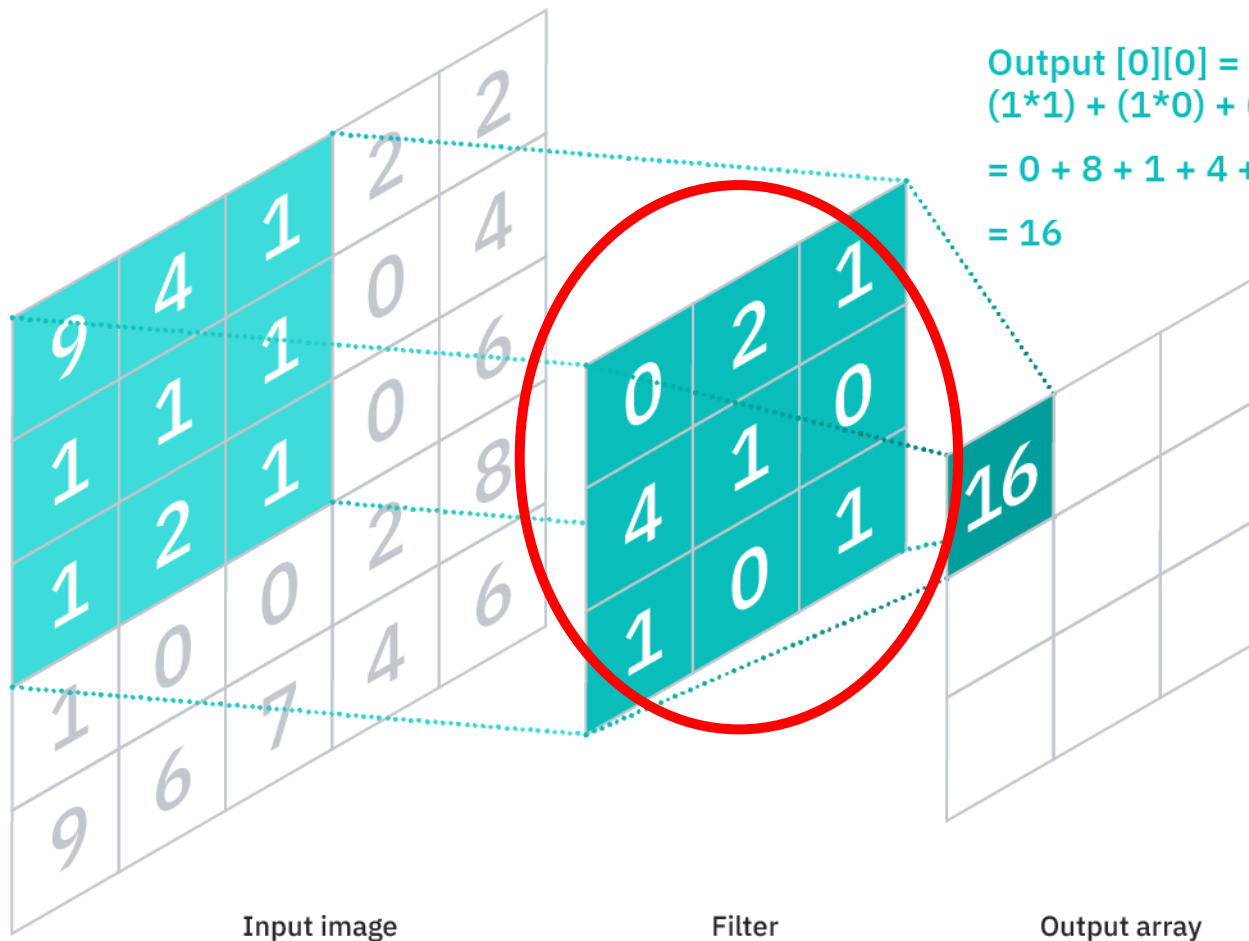
Convolutional neural networks



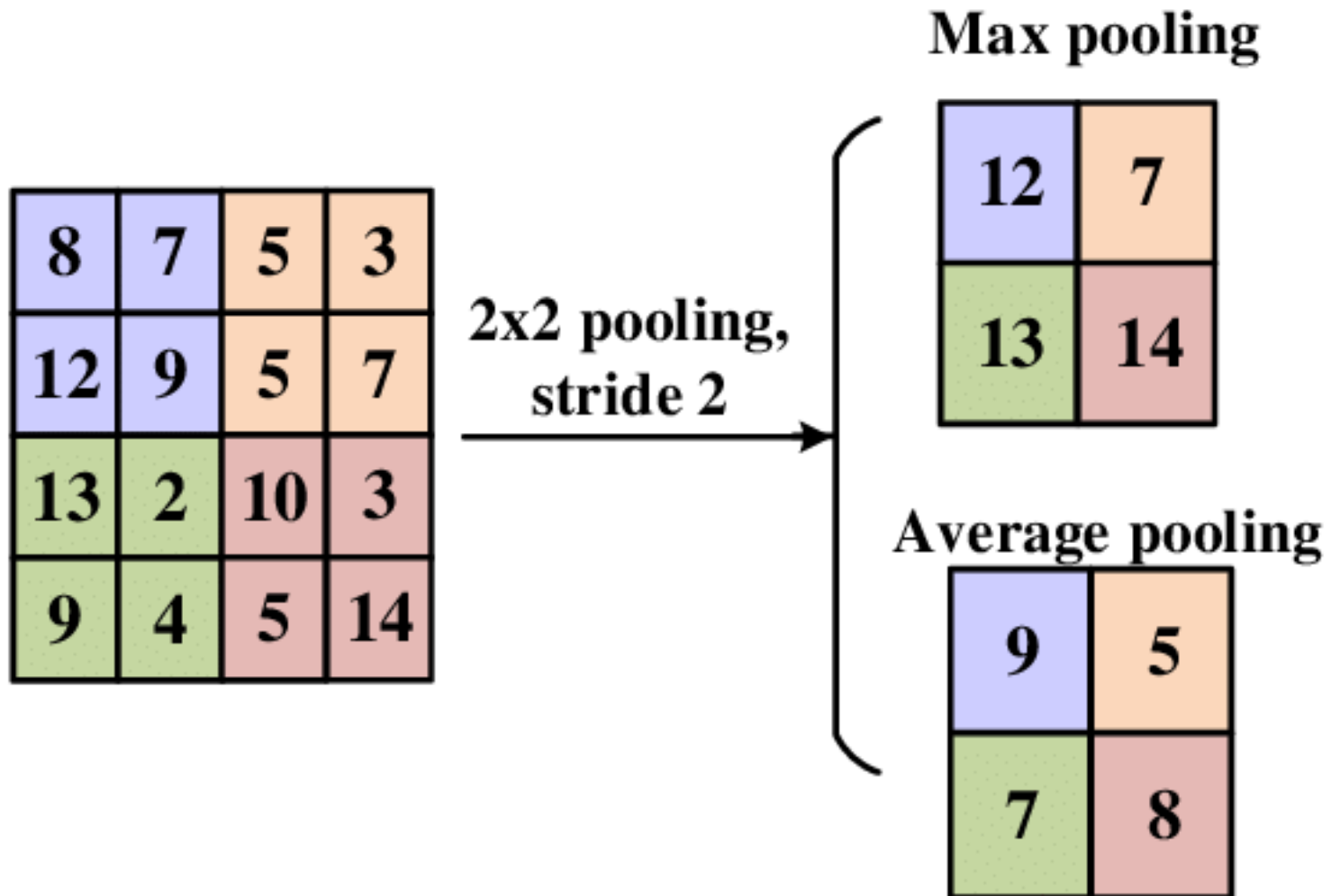
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Convolutional layer(s)



Pooling layer(s)



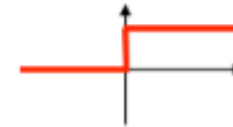
Activation functions



Unit step
(Heaviside)

$$\phi(z) = \begin{cases} 0, & z < 0, \\ 0.5, & z = 0, \\ 1, & z > 0, \end{cases}$$

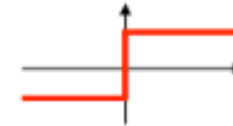
Perceptron
variant



Sign (Signum)

$$\phi(z) = \begin{cases} -1, & z < 0, \\ 0, & z = 0, \\ 1, & z > 0, \end{cases}$$

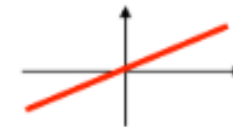
Perceptron
variant



Linear

$$\phi(z) = z$$

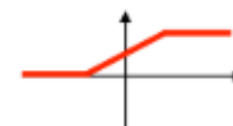
Adaline, linear
regression



Piece-wise linear

$$\phi(z) = \begin{cases} 1, & z \geq \frac{1}{2}, \\ z + \frac{1}{2}, & -\frac{1}{2} < z < \frac{1}{2}, \\ 0, & z \leq -\frac{1}{2}, \end{cases}$$

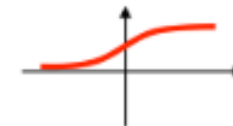
Support vector
machine



Logistic (sigmoid)

$$\phi(z) = \frac{1}{1 + e^{-z}}$$

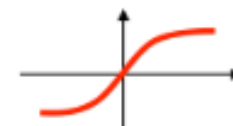
Logistic
regression,
Multi-layer NN



Hyperbolic tangent

$$\phi(z) = \frac{e^z - e^{-z}}{e^z + e^{-z}}$$

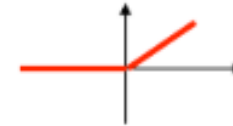
Multi-layer
Neural
Networks



Rectifier, ReLU
(Rectified Linear
Unit)

$$\phi(z) = \max(0, z)$$

Multi-layer
Neural
Networks



How to «learn»?



- Minimizing a «cost» function defined by the system's output and (indirectly) by the system's parameters

- Minimum Square Error (linear regression)

$$MSE = \frac{1}{2} \sum_{i=1}^N (y_i - \hat{y}_i)^2$$

- Cross Entropy (classification)

$$CE = - \sum_{i=1}^N y_i \cdot \log(\hat{y}_i)$$

- Methods: backpropagation

Deep network list



- Multilayer Perceptrons (MLPs)
- Convolutional Neural Networks (CNNs)
- Long Short Term Memory Networks (LSTMs)
- Recurrent Neural Networks (RNNs)
- Radial Basis Function Networks (RBFNs)
- Deep Belief Networks (DBNs)
- Restricted Boltzmann Machines(RBMs)
- Autoencoders (AE)
- Generative Adversarial Networks (GANs)
- Self Organizing Maps (SOMs)
- ...

That's all for today

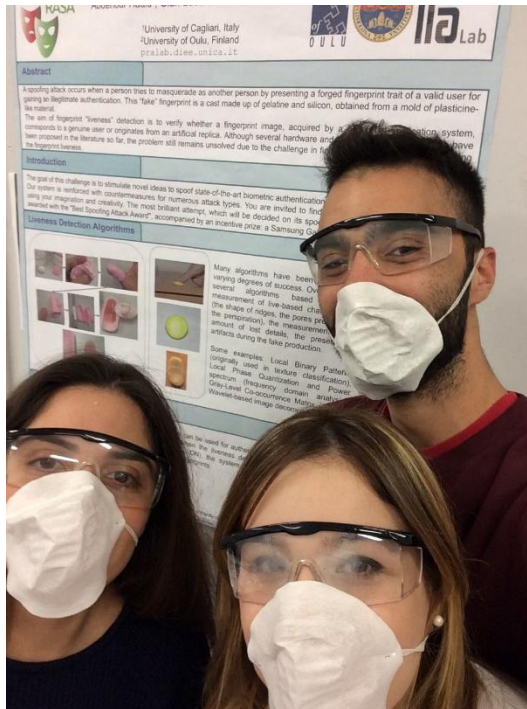


- **Computer vision: what is it?**
 - Definition
 - Image formation and processing
 - Image alignment and stitching
 - Depth and motion estimation
- **Some recalls about statistics**
 - Random variables
 - Distributions
 - Hypothesis verification tests
- **Don't miss the next exciting lecture!**





Thank you for listening!



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